Quantification: 877_wt_z087c1-2
==============================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	12
interior lacuna count	8
frame-edge lacuna count	4

Interior lacunae only: mean, median and sample SD (n = 8)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	4121.67	4349	1874.68	px^2
major_axis_length_px	lacuna shape	109.759	118.557	22.0224	px
minor_axis_length_px	lacuna shape	50.0334	51.7906	15.8766	px
aspect_ratio	lacuna shape	2.3302	2.1036	0.5642	unitless
eccentricity	lacuna shape	0.891	0.8795	0.0377	unitless
solidity	lacuna shape	0.7983	0.8195	0.0779	unitless
roots_count	headline	11.5	13	7.329	count
ring_length_r30_px	headline	383.5	400.5	220.995	px
ring_length_r60_px	reported	940.875	1105	500.73	px
owned_length_px	ownership-dependent	1955.13	2128.34	1279.16	px
edge_count	ownership-dependent	93.875	111	62.1138	count
mean_edge_length_px	ownership-dependent	20.7274	20.5054	2.07	px
perimeter_px	normalised	326.178	335.869	106.391	px
ring_area_r30_px2	normalised	10933.1	11032	2659.36	px^2
ring_area_r60_px2	normalised	26665	27035	5699.49	px^2
in_frame_fraction_r30	normalised	0.9993	1	0.002	unitless
in_frame_fraction_r60	normalised	0.9826	1	0.0493	unitless
ring_density_r30	normalised	0.0330377	0.0378335	0.0149541	px^-1
ring_density_r60	normalised	0.0334097	0.0390407	0.0159625	px^-1
roots_per_100px_perimeter	normalised	3.3601	3.798	1.9841	per 100 px
ring_attached_length_r30_px	network v2	342.5	366	225.096	px
ring_attached_length_r60_px	network v2	767	879	481.389	px
ring_length_w_r30_px	network v2	441.562	455.611	255.009	px
ring_length_w_r60_px	network v2	1081.41	1269.69	580.843	px
sholl_crossings_r10	network v2	9.25	10	5.2847	count
sholl_crossings_r20	network v2	12.125	11	6.8544	count
sholl_crossings_r30	network v2	13.75	15	6.6494	count
loop_count	loops, report only	8.5	11	6.8661	count
loop_edge_share	loops, report only	0.2327	0.3051	0.177	share
ring_width_mean_r30_px	width	6.9027	6.9112	0.8585	px

Per field
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measure	value	units
analysed_area_px2	1.009e+06	px^2
total_skeleton_length_px	30977	px
canalicular_length_density_per_px	0.0307008	px^-1
junction_count	630	count
junction_density_per_px2	0.00062438	px^-2
lacuna_count	12	count

lacunae_per_px2	1.189e-05	px^-2
skeleton_component_count	391	count
mean_component_length_px	79.2251	px
median_component_length_px	23	px
field_length_density_w_per_px	0.0350975	px^-1
field_density_without_flagged_per_px	0.0307008	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	716	px
unattached_component_count	36	count
canalicular_length_density_attached_per_px	0.0307008	px^-1
field_length_density_with_unattached_per_px	0.0314105	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	87	count
loop_edge_share	0.1932	share
edge_count	1615	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.3231	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.198	px
width_pixels_used	30977	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1757	77.0533	35.9762	2.1418	0.8843	0.7142	1.417	13.6949	531.608	True
2	4317	116.364	50.6241	2.2986	0.9004	0.8545	0.8274	37.7415	814.323	True
3	1047	73.2742	21.411	3.4223	0.9564	0.675	0.4727	102.495	237.658	False
4	5980	133.077	65.0796	2.0448	0.8723	0.7429	0.7964	208.483	500.988	False
5	4142	119.005	48.2749	2.4652	0.914	0.8826	0.2092	399.882	20.3518	True
6	4805	118.959	55.0153	2.1623	0.8866	0.8357	0.4696	413.392	307.567	False
7	3114	93.6668	46.7845	2.0021	0.8663	0.8167	0.8128	455.325	614.1	False
8	5891	121.42	63.3439	1.9168	0.8531	0.8972	1.4767	461.976	812.6	False
9	1692	71.502	34.0722	2.0985	0.8792	0.9082	0.0325	636.572	12.6684	True
10	3893	118.155	48.5658	2.4329	0.9116	0.8222	0.9667	739.935	833.772	False
11								860.708	550.499	False
12								970.308	685.063	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	9	343	732	1267.79	70	18.1112	7.5173
2	8	245	592	942.034	39	24.1547	7.215
3	13	374	1029	1888.16	97	19.4656	8.1759
4	23	761	1623	3410.19	164	20.7938	7.6347
5	5	215	584	438.262	28	15.6522	6.7369

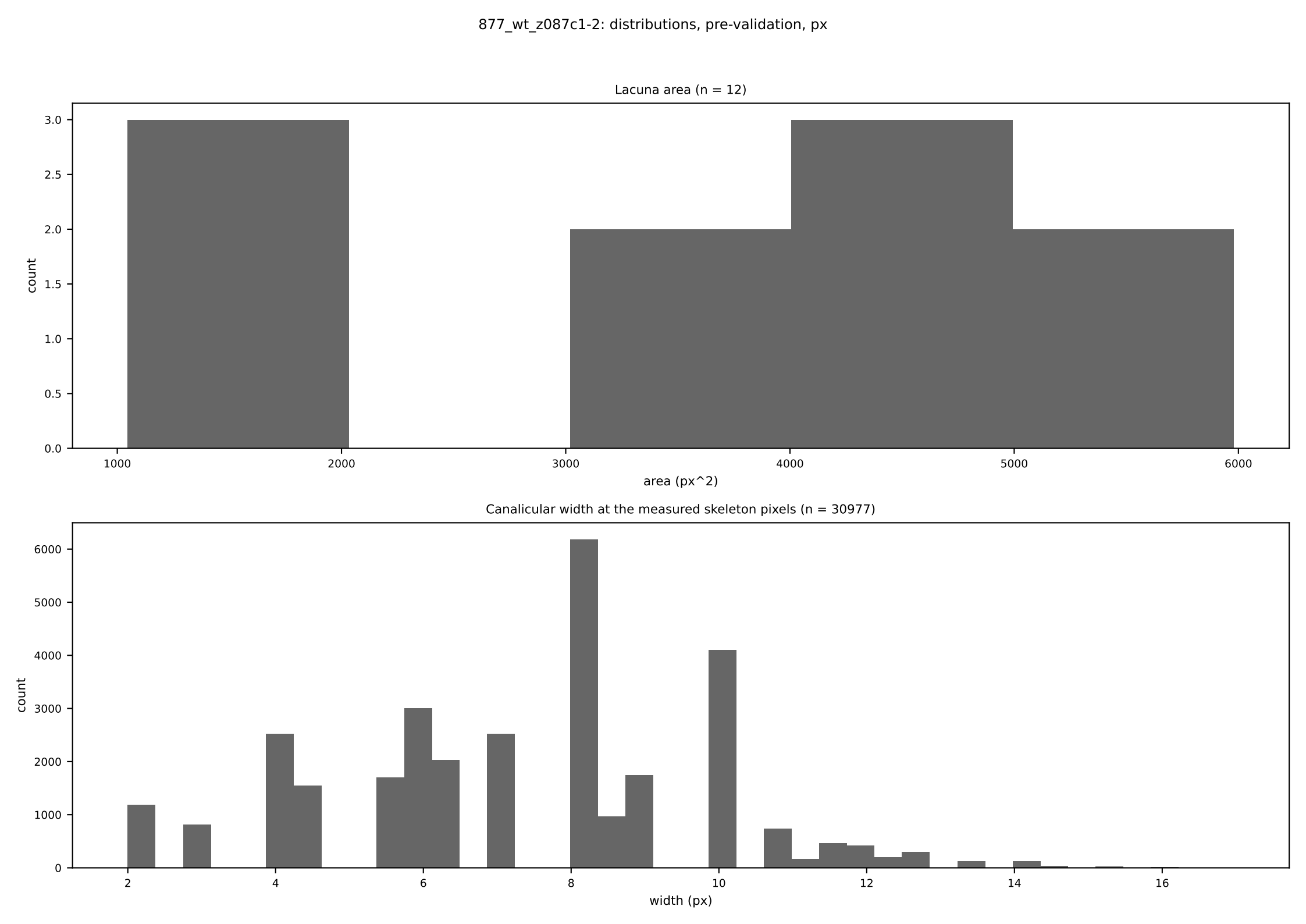
6	13	468	1125	2341.59	125	18.7327	7.2639
7	12	394	1120	2903.15	142	20.4447	6.7385
8	14	494	1090	3040.56	140	21.7183	7.0839
9	6	136	381	316.865	13	24.3742	7.9135
10	15	407	1154	1915.08	76	25.1985	6.5761
11	1	108	315	123.397	6	20.5662	6.4177
12	1	62	71	18.8995	1	18.8995	5.3312

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	274.066	6033	14351	0.6294	0.5895	0.056854	0.0510069	3.2839
2	340.475	9033	20414	0.7674	0.7052	0.0271228	0.0289997	2.3497
3	206.551	8300	22062	1	1	0.0450602	0.0466413	6.2938
4	502.186	15464	35653	1	1	0.0492111	0.0455221	4.58
5	336.628	7317	17241	0.6107	0.5856	0.0293836	0.0338728	1.4853
6	357.404	12357	30057	1	1	0.0378733	0.0374289	3.6373
7	314.333	10425	25683	1	1	0.0377938	0.0436086	3.8176
8	364.718	12512	29886	1	1	0.0394821	0.0364719	3.8386
9	190.953	4836	12429	0.5823	0.5596	0.0281224	0.0306541	3.1421
10	397.002	11639	28387	1	1	0.0349686	0.0406524	3.7783
11	301.404	9757	24715	1	1	0.011069	0.0127453	0.3318
12	165.823	7011	16877	0.9943	0.8606	0.00884325	0.00420691	0.6031

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	311	618	395.363	846.252	8	7	10
2	227	455	289.492	684.896	8	9	11
3	349	965	422.777	1160.41	9	10	14
4	734	1490	878.394	1882.16	16	21	22
5	153	321	240.823	658.742	6	6	7
6	413	930	554.441	1309.78	13	18	19
7	332	874	448.99	1289.76	10	11	16
8	448	869	565.943	1249.63	13	19	18
9	134	281	154.711	424.848	5	4	7
10	383	884	462.233	1324.11	10	11	13
11	64	107	130.024	358.333	2	6	4
12	17	17	69.6985	77.1127	1	1	4


877_wt_z087c1-2: distributions, pre-validation, px
Lacuna area (n = 12)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 30977)
count
width (px)